IGCSE 0607 Practice Questions — E10.2 — Combined Questions & Markscheme

Interpreting statistical data

Total Questions: 18 — **Total Marks:** 118 Marks

Table of Contents & Quick Navigation

- **E10.2** Interpreting statistical data
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(10 Qs)*

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1 The table shows information about the marks scored by two different classes, Class A and Class B, in a mathematics examination.

Statistic	Class A	Class B
Mean	62	70
Median	65	72
Range	40	25
Interquartile Range	15	10

(a) Which class had the higher average mark?
Give a reason for your answer.

..... **[1]**

(b) Which class had marks that were more consistent?
Give a reason for your answer.

..... **[1]**

(c) A student says, "Every student in Class B scored a higher mark than every student in Class A."
Explain why this statement cannot be confirmed using only the information in the table.

.....

.....

..... **[2]**

2 The number of goals scored by Team A in each of their last 6 matches is recorded below.

3 1 0 4 2 2

Team B also played 6 matches.
For Team B, the median number of goals scored was 1.5 and the range was 5.

(a) Work out the median and the range of the number of goals scored by Team A.

Median = ... , Range = ... **[2]**

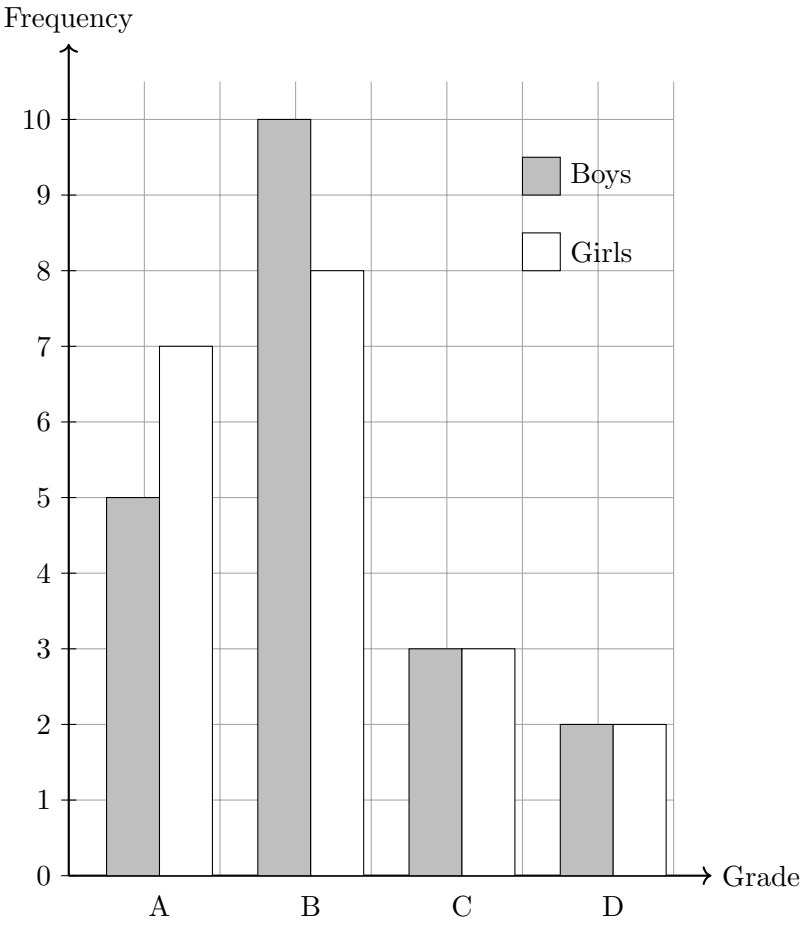
(b) Make two comments comparing the performance of Team A with the performance of Team B.

.....

.....

..... **[2]**

3 The dual bar chart shows the grades achieved by a group of boys and a group of girls in an examination.



(a) Write down the modal grade for the boys.

..... **[1]**

(b) Find the number of girls who scored Grade C.

..... **[1]**

(c) Calculate the percentage of all students who scored Grade A.

..... % **[2]**

4 The stem-and-leaf diagram shows the heights, in centimetres, of 15 plants.

1	2 5 8
2	0 1 1 4 7
3	2 3 5 9
4	0 4
5	1

Key: 1 | 2 represents 12 cm

(a) Find the median height.

..... cm **[1]**

(b) Find the interquartile range of the heights.

..... cm **[2]**

(c) Find the probability that a plant chosen at random has a height greater than 30 cm.

..... **[1]**

5 The stem-and-leaf diagram shows the ages, in years, of 15 members of a chess club.

1	2 5 8
2	0 1 1 4 7 9
3	2 5 6 9
4	1 3

Key: 2 | 1 represents 21 years

(a) Find the median age.

..... years **[1]**

(b) Find the range of the ages.

..... years **[1]**

(c) Find the interquartile range.

..... years **[2]**

(d) A member is chosen at random.
Find the probability that they are older than 30 years.

..... **[1]**

6 The back-to-back stem-and-leaf diagram shows the scores, out of 50, achieved by 15 students in Class A and 15 students in Class B in a mathematics test.

Class A	$tem	Class B
9 8	1	4 6 8
8 7 5 3	2	0 2 4 5 9
9 6 4 2 1	3	0 3 5 8
7 5 2 0	4	1 6 9

Key: 3 | 2 | 0 represents 23 for Class A and 20 for Class B

(a) Write down the median score and the interquartile range for Class A.

Median = ... , Interquartile range = ... **[2]**

(b) For Class B, the median score is 29 and the interquartile range is 18.

Compare the test scores of Class A and Class B. Make two comparative statements, one about average performance and one about the spread of the scores.

.....

..... **[2]**

(c) A student claims: "More students in Class B scored above 40 than in Class A." Explain whether the diagram supports this claim.

..... **[2]**

(d) Explain one restriction on using these results to conclude that Class A is better at mathematics than Class B.

..... **[1]**

7 A set of 5 integers has the following statistical properties:

- Mode = 3
- Median = 7
- Mean = 8
- Range = 11

(a) Find the 5 integers.

..... **[4]**

(b) A sixth integer, x, is added to the set.
The new mean is 9.
Work out the value of x.

$x =$ ..... **[2]**

(c) A student claims: "If another completely different set of 5 integers has the same range of 11, it must also have the same standard deviation."

State whether the student is correct.
Give a statistical reason for your answer.

.....

..... **[1]**

8 Two branches of a call centre, Branch North and Branch South, record the duration, t minutes, of customer support calls during one morning.
The table shows the distribution of call durations for both branches.
Branch North recorded 80 calls in total.
In Branch South, the relative frequency of a call lasting more than 8 minutes is $\frac{13}{20}$.

Call duration (t minutes)	Branch North	Branch South
$0 < t \leq 4$	12	15
$4 < t \leq 8$	28	x
$8 < t \leq 12$	y	35
$12 < t \leq 16$	14	20
$16 < t \leq 20$	6	10

(a) Find the value of y and the value of x.

$y =$ ..... , $x =$ ..... **[4]**

(b) Calculate an estimate of the mean call duration for Branch North and for Branch South.

Branch North = ... , Branch South = ... **[4]**

(c) A supervisor inspects the table and makes two statements:

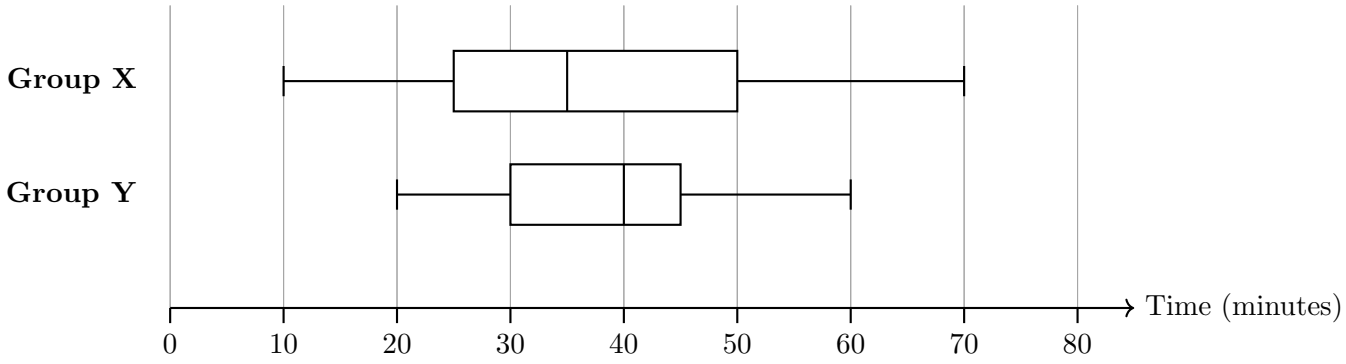
- Statement 1: "The maximum possible range of call durations in Branch North is 20 minutes."
- Statement 2: "At least 18 calls in Branch South lasted between 9 minutes and 11 minutes."

Explain whether each statement is valid, giving a reason based on the grouped data.

.....

..... **[4]**

9 The box-and-whisker plots show information about the times, in minutes, taken by two groups of people (Group X and Group Y) to complete a puzzle.



(a) Find the median time for Group Y.

..... minutes **[1]**

(b) Find the interquartile range for Group X.

..... minutes **[1]**

(c) Make two different comments comparing the distributions of the times taken by Group X and Group Y.

.....

.....

.....

..... **[2]**

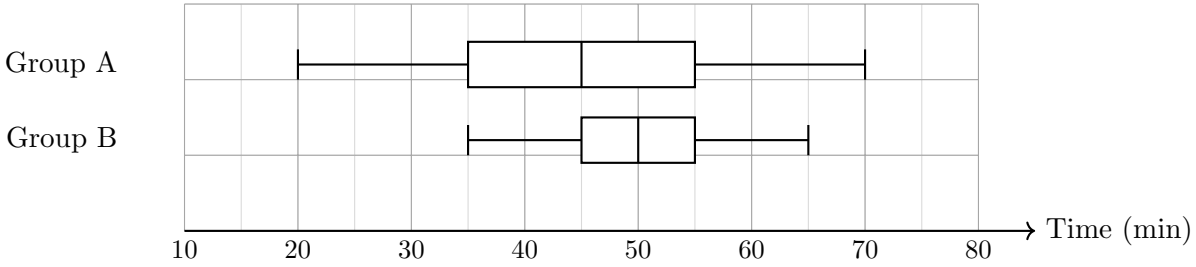
(d) Which group had a larger proportion of people completing the puzzle in less than 30 minutes? Explain your reasoning.

.....

.....

..... **[2]**

10 The box-and-whisker plots show information about the times, in minutes, taken by two groups of students, Group A and Group B, to complete a mathematics test.



(a) Write down the median time for Group A.

..... minutes **[1]**

(b) Work out the interquartile range for Group B.

..... minutes **[1]**

(c) Make two comments comparing the distributions of the times taken by Group A and Group B.

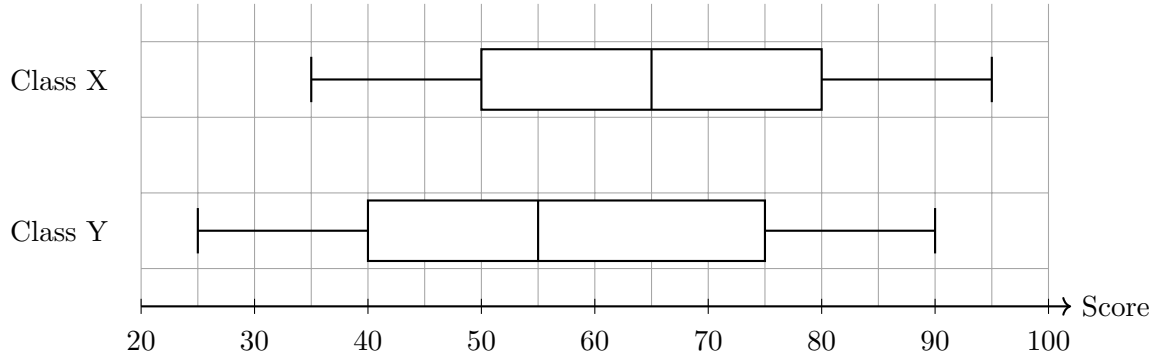
Comment 1:

Comment 2: **[2]**

(d) Group A has 80 students.
Estimate the number of students in Group A who took longer than 55 minutes.

..... students **[2]**

11 The box-and-whisker plots show the distribution of test scores for two classes, Class X and Class Y.



(a) Write down the median score for Class X.

..... **[1]**

(b) Work out the interquartile range for Class Y.

..... **[2]**

(c) Make two different comments comparing the distribution of test scores in Class X with Class Y.

[2]

(d) A student states: "Because the median score for Class X is higher, every student in Class X scored more than every student in Class Y."

Is this statement correct?
Give a statistical reason for your answer.

.....

..... **[1]**

12 The table shows information about the mass, m grams, of 60 mangoes collected by a farmer.

Mass (m grams)	$100 < m \leq 120$	$120 < m \leq 140$	$140 < m \leq 160$	$160 < m \leq 180$	$180 < m \leq 200$
Frequency	10	20	15	10	5

(a) State the modal class.

..... **[1]**

(b) Calculate an estimate of the mean mass of the mangoes.

..... g **[4]**

(c) The farmer states that exactly 15 mangoes have a mass of 150 grams because 150 is the midpoint of the class $140 < m \leq 160$.
Explain why the farmer's reasoning is incorrect.

.....

.....

..... **[2]**

13 The table shows the number of goals scored by a football team in each of 40 matches.

Number of goals	0	1	2	3	4	5
Frequency	8	12	x	6	y	2

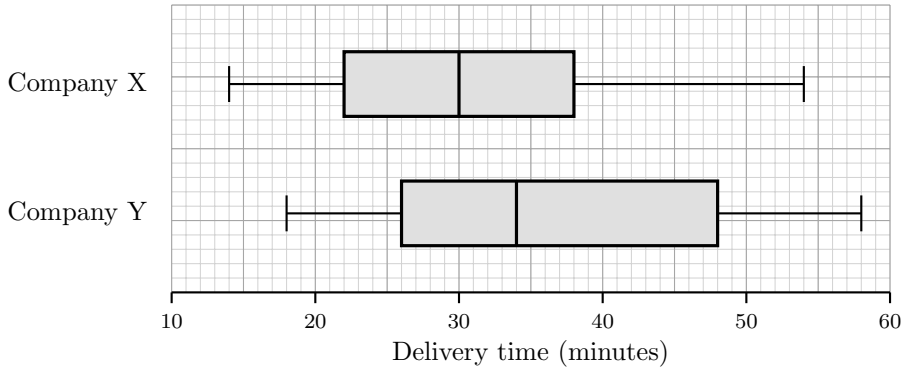
(a) The mean number of goals scored per match is 1.7.
Show that $2x + 4y = 30$.

[3]

(b) Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

14 The box-and-whisker plots show information about the delivery times, in minutes, of parcels delivered by two courier companies, Company X and Company Y.



(a) For Company Y, find

(i) the median delivery time,

..... minutes **[1]**

(ii) the interquartile range of the delivery times.

..... minutes **[2]**

(b) Company X recorded 160 deliveries and Company Y recorded 240 deliveries.

(i) Calculate the number of deliveries by Company Y that took 26 minutes or less.

..... **[2]**

(ii) A manager claims: "More deliveries took between 22 minutes and 38 minutes for Company X than for Company Y." Explain why it is not possible to be certain from the box-and-whisker plots whether this claim is true.

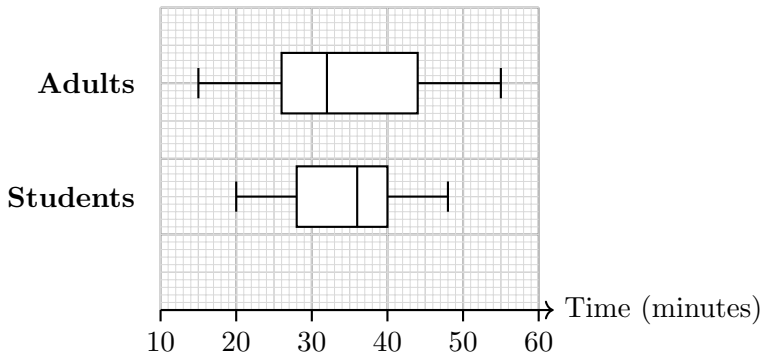
..... **[3]**

(c) A customer wants a courier company that is both faster on average and more consistent in its delivery times.
State which company the customer should choose, giving two statistical reasons to justify your decision.

.....

..... **[3]**

15 The box-and-whisker plots show information about the times, in minutes, taken by a group of Adults and a group of Students to complete a logic puzzle.



(a) Use the diagram to find the interquartile range of the times taken by the Adults.

..... minutes **[2]**

(b) Make two different statements comparing the distribution of the times taken by the Adults with the times taken by the Students.

.....

.....

..... **[2]**

(c) A student claims: "Because the range of the Students' times is smaller, this proves that every student completed the puzzle faster than the slowest Adult."

Comment on the validity of this reasoning.

.....

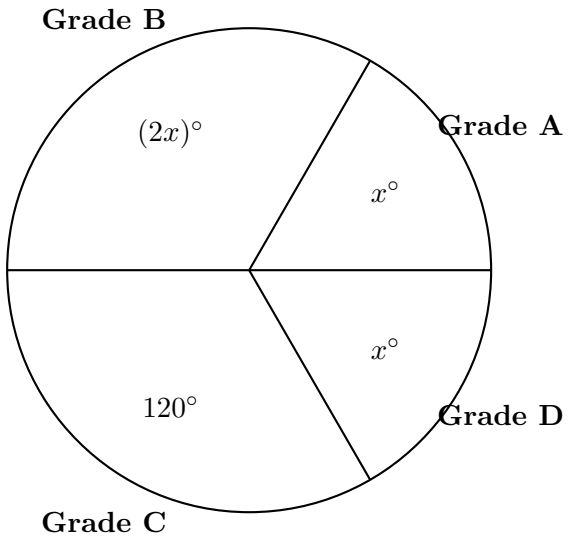
.....

..... **[2]**

16 The pie chart shows the distribution of grades (A, B, C, and D) achieved by 120 students in Class P.

The frequency table shows the distribution of the same grades achieved by the 80 students in Class Q.

The ratio of the number of students in Class Q achieving Grade A : Grade B : Grade C : Grade D is 1 : 3 : 2 : 2.



Class P

(a) Find the number of students who achieved Grade A in Class P.

..... **[3]**

(b) A student claims: "A higher percentage of students in Class Q achieved Grade B than in Class P."

Show that this claim is correct.

[3]

(c) The school principal states that Class P performed better overall because more students achieved Grade C in Class P than in Class Q.

Give one statistical reason why comparing the absolute frequencies (number of students) for

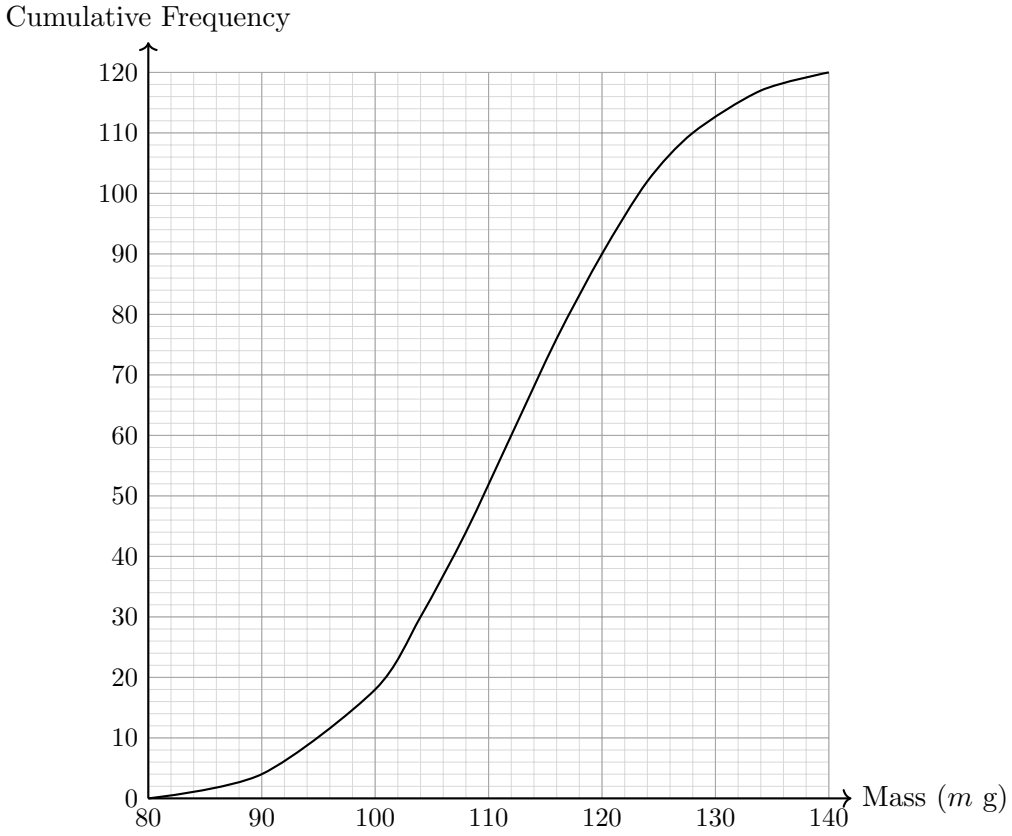
each grade is not an appropriate way to determine which class performed better.

.....

.....

..... **[1]**

17 The cumulative frequency curve shows information about the masses, m grams, of 120 apples from Farm X.



(a) Use the graph to find an estimate for the median mass.

..... g **[1]**

(b) Use the graph to find an estimate for the interquartile range.

..... g **[2]**

(c) Apples with a mass greater than 128 g are classed as Large.
Find the percentage of these apples that are Large.

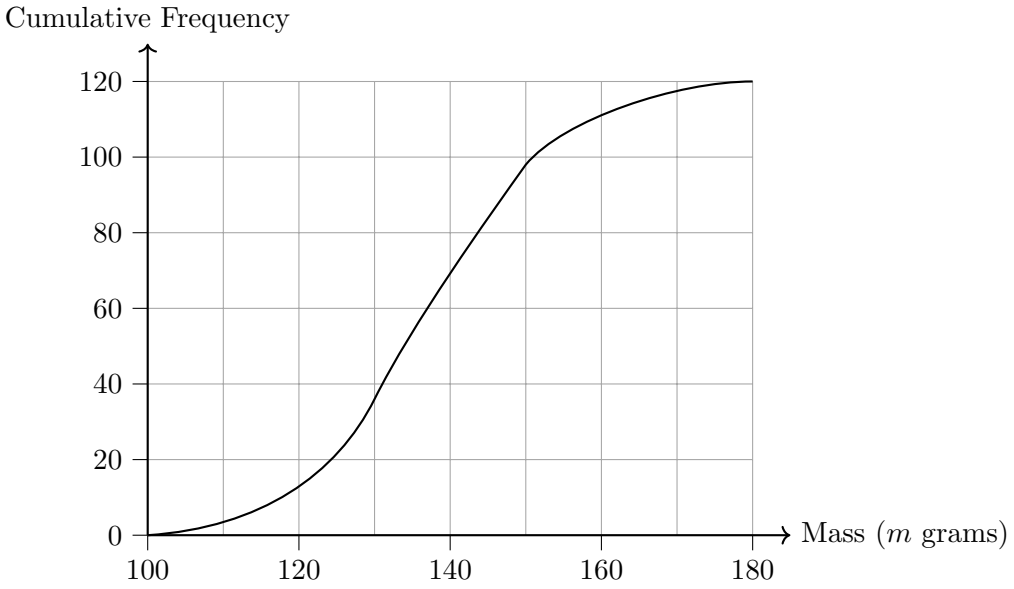
..... % **[2]**

(d) 120 apples from Farm Y have a median mass of 105 g and an interquartile range of 22 g.
Compare the distribution of the masses of the apples from Farm X and Farm Y.

.....

..... [2]

18 The cumulative frequency graph shows information about the masses, m grams, of 120 apples.



(a) Use the graph to find an estimate for

(i) the median,

..... g **[1]**

(ii) the interquartile range.

..... g **[2]**

(b) Apples with a mass greater than 150 grams are classified as 'large'.
Find an estimate for the number of large apples.

..... **[2]**

(c) Two of the 120 apples are chosen at random.
Find the probability that both apples have a mass of 150 grams or less.

..... **[3]**

1 **(a)** Class B, because the mean (or median) is higher. **[B1]**

(b) Class B, because the range (or interquartile range) is smaller. **[B1]**

(c) Identifies that the table only gives summary statistics / does not show individual marks **[B1]**

Mentions overlapping ranges / a specific example (e.g. the lowest mark in Class B could be lower than the highest mark in Class A) **[B1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** Median = 2 **[B1]**

Range = 4 **[B1]**

(b) Team A generally scored more goals because their median is higher ($2 > 1.5$) **[B1]**

Team A was more consistent because their range is smaller ($4 < 5$) **[B1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** B **[B1]**

(b) 3 **[B1]**

(c) Total students $= 5+10+3+2+7+8+3+2 = 40$ AND Total Grade A $= 5+7 = 12$ **[M1]**

30 **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** 27 **[B1]**

(b) Upper quartile = 35 OR Lower quartile = 20 **[M1]**

15 **[A1]**

(c) $\frac{7}{15}$ **[B1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** 27 **[B1]**

(b) 31 **[B1]**

(c) 36 and 20 identified as quartiles **[M1]**

16 **[A1]**

(d) $\frac{6}{15}$ or $\frac{2}{5}$ **[B1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** Median = 32 **[B1]**

Interquartile range $= 40 - 25 = 15$ **[B1]**

(b) Class A has a higher median score ($32 > 29$), so on average Class A performed better than Class B **[B1]**

Class A has a smaller interquartile range ($15 < 18$), so Class A's scores are more consistent (less spread out) than Class B's **[B1]**

(c) In Class B, 3 students scored above 40 ($41, 46, 49$), and in Class A, 3 students also scored above 40 ($42, 45, 47$, as 40 is not above 40) **[M1]**

No, the claim is incorrect because both classes had the same number (3 students) scoring above 40 **[A1]**

(d) The results are from only one test (or a small sample of 15 students per class) and may not represent overall mathematical ability across all topics **[B1]**

[Back to Question] *[Total Marks: 7]*

7 (a) Identifies the middle value is 7 **[M1]**

Identifies that 3 must appear at least twice as the lowest values: $3, 3$ **[M1]**

Uses the range to find the maximum value: $3 + 11 = 14$ **[M1]**

Uses the mean to find the missing value: $40 - (3 + 3 + 7 + 14) = 13$, yielding $3, 3, 7, 13, 14$ **[A1]**

(b) New sum $= 6 \times 9 = 54$ AND Old sum $= 5 \times 8 = 40$ (or $54 - 40$) **[M1]**

14 **[A1]**

(c) No, because standard deviation measures the spread of all data points around the mean, not just the difference between the extreme values **[B1]**

[Back to Question] *[Total Marks: 7]*

8 (a) $y = 80 - (12 + 28 + 14 + 6) = 20$ **[B1]**

Calls lasting more than 8 minutes in Branch South $= 35 + 20 + 10 = 65$, giving $\dfrac{65}{80 + x} = \dfrac{13}{20}$ **[M1]**

$13(80 + x) = 1300$ giving $80 + x = 100$ **[M1]**

$x = 20$ **[A1]**

(b) Branch North: $\sum fm = 12(2) + 28(6) + 20(10) + 14(14) + 6(18) = 696$ **[M1]**

Branch North estimated mean $= \dfrac{696}{80} = 8.7$ minutes **[A1]**

Branch South: $\sum fm = 15(2) + 20(6) + 35(10) + 20(14) + 10(18) = 960$ **[M1]**

Branch South estimated mean $= \dfrac{960}{100} = 9.6$ minutes **[A1]**

(c) Statement 1 is valid because the shortest possible call duration is just above 0 minutes and the longest possible duration is 20 minutes, giving a maximum possible range of $20 - 0 = 20$ minutes **[B2]**

Statement 2 is not valid because the 35 calls in the interval $8 < t \leq 12$ are grouped, so all

35 calls could lie outside $9 \leq t \leq 11$ (for example, between 8 and 9 minutes) **[B2]**

[Back to Question] *[Total Marks: 12]*

9 **(a)** 40 **[B1]**

(b) 25 (from $50 - 25$) **[B1]**

(c) Group Y was slower on average / had a higher median time ($40 > 35$) **[B1]**

Group Y's times were more consistent / had a smaller IQR ($15 < 25$) or smaller range ($40 < 60$) **[B1]**

(d) Identifies that exactly 25% of Group Y took less than 30 minutes (since lower quartile is 30) **[M1]**

States Group X had a larger proportion, because 30 is between their LQ (25) and median (35), meaning more than 25% took less than 30 minutes **[A1]**

[Back to Question] *[Total Marks: 6]*

10 **(a)** 45 **[B1]**

(b) 10 **[B1]**

(c) Group A students were faster on average because their median time (45 min) is less than Group B (50 min). **[B1]**

Group B times are more consistent / less spread out because their interquartile range (10 min) is smaller than Group A (20 min). **[B1]**

(d) Recognises that 55 is the upper quartile, representing 25% or $\frac{1}{4}$ of the students **[M1]**

20 **[A1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** 65 **[B1]**

(b) $75 - 40$ **[M1]**

35 **[A1]**

(c) Class X has a higher average score (or median is higher: $65 > 55$) **[B1]**

Class X scores are less spread out / more consistent (or IQR is lower: $30 < 35$, or Range is lower: $60 < 65$) **[B1]**

(d) No, because the ranges overlap (e.g., the maximum score in Class Y is 90, which is higher than the lower quartile of Class X) **[B1]**

[Back to Question] *[Total Marks: 6]*

12 **(a)** $120 < m \leq 140$ **[B1]**

(b) Use of correct midpoints: $110, 130, 150, 170, 190$ **[M1]**

$110 \times 10 + 130 \times 20 + 150 \times 15 + 170 \times 10 + 190 \times 5$ **[M1]**

$= 1100 + 2600 + 2250 + 1700 + 950 = 8600$ **[M1]**

$\frac{8600}{60} = 143.33\ldots$ so 143 (to 3 sf) **[A1]**

(c) The exact masses of the individual mangoes are unknown because the data is grouped **[B1]**

The 15 mangoes could have any combination of masses between 140 g and 160 g, not necessarily exactly at the midpoint **[B1]**

[Back to Question] *[Total Marks: 7]*

13 (a) Sum of goals $= (0 \times 8) + (1 \times 12) + (2 \times x) + (3 \times 6) + (4 \times y) + (5 \times 2)$ OR $12 + 2x + 18 + 4y + 10$ **[M1]**

Equates sum of goals to 40×1.7 OR $\frac{40 + 2x + 4y}{40} = 1.7$ **[M1]**

$40 + 2x + 4y = 68$

2x + 4y = $28(Correctcompletionwithnoerrors)[A1]$

(b) $8 + 12 + x + 6 + y + 2 = 40$ **[M1]**

$x + y = 12$ **[A1]**

Solves simultaneous equations: e.g. $2x + 4(12 - x) = 28$

2x + 48 - 4x = 28

$2x = 20[M1]$

$x = 10, y = 2$ **[A1]**

[Back to Question] *[Total Marks: 7]*

14 (a) (i) 34 **[B1]**

(ii) $48 - 26$ **[M1]**

22 **[A1]**

(b) (i) 0.25×240 **[M1]**

60 **[A1]**

(ii) For Company X, 22 and 38 are the lower and upper quartiles, so 50% of 160 = 80 deliveries took between 22 and 38 minutes **[B1]**

For Company Y, 22 and 38 are not quartiles, so the exact proportion of deliveries in this interval cannot be determined from the box plot **[B1]**

Since Company Y had more total deliveries (240 > 160), Company Y could have more than 80 deliveries in that interval **[B1]**

(c) Company X **[B1]**

Company X has a lower median delivery time (30 minutes < 34 minutes), so it is faster on average **[B1]**

Company X has a smaller interquartile range (16 minutes < 22 minutes), so its delivery times are more consistent **[B1]**

[Back to Question] *[Total Marks: 11]*

15 **(a)** Upper Quartile = 44 and Lower Quartile = 26 **[M1]**

$44 - 26 = 18$ **[A1]**

(b) The Adults generally completed the puzzle faster because their median time is lower ($32 < 36$) **[B1]**

The Students' times were more consistent (less spread out) because their interquartile range (12) is smaller than the Adults' interquartile range (18) **[B1]**

(c) The reasoning is invalid because range only measures spread, not the actual values or maximum times **[B1]**

However, the conclusion happens to be true because the maximum time for Students (48) is less than the maximum time for Adults (55) **[B1]**

[Back to Question] *[Total Marks: 6]*

16 **(a)** Forms a correct equation: $x + 2x + 120 + x = 360$ **[M1]**

Solves to find $4x = 240$

x = 60$[M1]$

Calculates number of students: $\frac{60}{360} \times 120 = 20$ **[A1]**

(b) Calculates the percentage for Class P: $\frac{120^\circ}{360^\circ} \times 100 = 33.3\%$ (or $\frac{1}{3}$) **[M1]**

Identifies total ratio parts for Class Q $= 1 + 3 + 2 + 2 = 8$ **[M1]**

Calculates percentage for Class Q: $\frac{3}{8} \times 100 = 37.5\%$ and concludes $37.5\% > 33.3\%$ **[A1]**

(c) The two classes have different total numbers of students (so proportions/percentages must be used for a fair comparison). **[B1]**

[Back to Question] *[Total Marks: 7]*

17 **(a)** 112 **[B1]**

(b) 120 and 104 seen as quartiles **[M1]**

16 **[A1]**

(c) Cumulative frequency of 110 seen, leading to $120 - 110 = 10$ apples **[M1]**

8.33 (8.333...) **[A1]**

(d) Farm X apples are heavier on average as their median (112 g) is greater than Farm Y (105 g). **[B1]**

Farm X apples are more consistent in mass / less spread out as their interquartile range (16 g) is smaller than Farm Y (22 g). **[B1]**

[Back to Question] *[Total Marks: 7]*

18 **(a)** **(i)** 136 to 138 **[B1]**

(ii) Upper quartile (90th) = 146 to 148 AND Lower quartile (30th) = 126 to 128 **[M1]**

18 to 22 **[A1]**

(b) Reads cumulative frequency at 150g as 98 (allow 97 to 99) **[M1]**

$120 - 98 = 22$ (allow 21 to 23) **[A1]**

(c) Number of apples $\leq$ 150g is 98 (or their read value from (b)) **[B1]**

$\frac{98}{120} \times \frac{97}{119}$ **[M1]**

$\frac{9506}{14280}$ or $\frac{4753}{7140}$ or 0.666 $(0.6656\ldots)$ **[A1]**

[Back to Question] *[Total Marks: 8]*